ETL Health Colombia — ODS 3: Dimensional Data Warehouse for Healthcare Analytics

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1 Introduction

1.1 Problem Definition

Colombia's healthcare system (SGSSS) faces significant territorial disparities in access to services. While the country has achieved high affiliation rates nationally, the distribution of healthcare infrastructure (beds, procedural rooms, equipment) does not proportionally match the population's needs across departments and municipalities. Some regions maintain high capacity per affiliate while others face critical shortages.

This project builds a dimensional data warehouse that integrates two complementary datasets from the Colombian Ministry of Health:

1. **Affiliates (SISPRO):** Number of healthcare affiliates by department, municipality, and regime type (Subsidized, Contributory, Special, Individual).
2. **Facilities (REPS):** Installed healthcare infrastructure capacity (beds, rooms, equipment) by facility, department, and municipality.

Understanding the relationship between affiliation density and infrastructure capacity is essential for identifying underserved regions and informing resource allocation decisions.

1.2 SDG Alignment

This project is aligned with **UN Sustainable Development Goal 3 — Good Health and Well-Being**, specifically **Target 3.8: Achieve universal health coverage (UHC)**, including financial risk protection, access to quality essential healthcare services, and access to safe, effective, quality, and affordable essential medicines and vaccines for all.

The analysis directly addresses Target 3.8 by examining:

- **Coverage breadth** (R1, R2): Measuring how many Colombians are affiliated and where they are located.
- **Service availability** (R3, R4): Assessing whether healthcare infrastructure exists to serve the affiliated population.
- **Equity** (R5): Examining whether access is equitably distributed across regimes and territories.

1.3 Geographic and Population Scope

- **Geographic scope:** National — 33 departments + Bogotá D.C., across all municipalities.
- **Population of interest:** Affiliates of the Colombian healthcare system and healthcare facilities (IPS).
- **Potential stakeholders:** Ministry of Health and Social Protection (Minsalud), departmental health secretaries, EPS (healthcare promoting entities), hospital managers, healthcare policy researchers.

2 Analytical Objectives and Requirements

2.1 Analytical Objective

Analyze the distribution of healthcare affiliates and infrastructure capacity across Colombian territories to identify geographic disparities and support resource allocation decisions aligned with SDG Target 3.8.

2.2 Requirements Matrix (R1–R5)

ID	Analytical Requirement	Business Question	Decision Supported
R1	Determine the total number of affiliates by regime type nationally	How many Colombians are affiliated to each regime (Subsidized, Contributory, Special, Individual)?	Understand the composition of the healthcare system for policy planning.
R2	Identify the departments with the highest and lowest affiliate density	Which departments concentrate the most/least affiliates?	Detect territorial disparities to prioritize interventions.
R3	Analyze the distribution of healthcare facilities by nature (public/private) and care level	What is the public-private mix and at what care levels?	Evaluate whether the private sector complements public infrastructure.
R4	Compare installed capacity per affiliate across departments	Which departments have sufficient infrastructure relative to their affiliated population?	Identify regions where expansion is needed.
R5	Assess the relationship between regime type and geographic distribution	Are Subsidized affiliates concentrated in different territories than Contributory?	Understand whether regime segmentation maps to geographic inequity.

Table 1: Analytical requirements matrix.

3 Data Sources

3.1 Source Selection

Characteristic	Affiliates Dataset	Facilities Dataset
Source	SISPRO — Sistema Integrado de Información de la Protección Social	REPS — Registro Especial de Prestadores de Servicios de Salud
Owner	Ministerio de Salud y Protección Social	Ministerio de Salud y Protección Social
URL	datos.gov.co	datos.gov.co
Format	CSV (period as thousand separator)	CSV
Records	3,369 rows	41,427 rows (9,320 unique facilities)
Attributes	8 columns	20 columns
Temporal	April 2022 (Q2 snapshot)	November 2022 (Q4 snapshot)
Measures	`NumPersonas` (affiliates)	`num cantidad capacidad instalada` (capacity)

Table 2: Data source characteristics.

3.2 Dataset Suitability

Both datasets are suitable for dimensional modeling:

- Clear grain: affiliates per (department, municipality, regime, month); facilities per (provider, capacity type).
- Natural dimensions: time, geography, regime, facility, capacity type.
- Official open data from Minsalud, publicly available on datos.gov.co.

4 Data Profiling and Quality Assessment

4.1 Affiliates Dataset

Attribute	Nulls	Unique	Notes
CodDepto	0	34	Department DANE code
Departamento	0	34	Department name
CodMunicipio	0	1,046	Municipality DANE code
Municipio	0	1,046	Municipality name
IDRegimen	0	4	S/C/E/I
Ano	0	1	2022
Mes	0	1	4 (April only)
NumPersonas	0	—	Range: 1–6,410,877. Sum: 51,182,238

Table 3: Affiliates dataset profiling.

Quality issues:

- `NumPersonas` stored as string with period thousand separators (e.g., "2.194" = 2,194).
- Temporal coverage limited to single month (April 2022).
- 2 rows with `Departamento = ''NO APLICA''` containing 1,315,701 affiliates (57% of Especial regime, 96% of Individual regime).

4.2 Facilities Dataset

Attribute	Nulls	Unique	Notes
Departamento	0	38	Includes special health districts
Municipio	0	—	Municipality name
Codigo prestador	0	9,320	Unique provider identifier
naturaleza	0	3	Publica, Privada, Mixta
num nivel atencion	25,266 (61%)	—	Care level (1–4)
nom grupo capacidad	0	—	CAMAS, SALAS, etc.
num cantidad capacidad	0	—	Range: 1–650+

Table 4: Facilities dataset profiling.

Quality issues:

- 61% null values in `num nivel atencion` (care level).

- Each facility appears in multiple rows (one per capacity type entry).
- 5 special health districts (Cali, Cartagena, Barranquilla, Santa Marta, Buenaventura) reported as "department" instead of their real department.

5 Dimensional Data Model

5.1 Star Schema

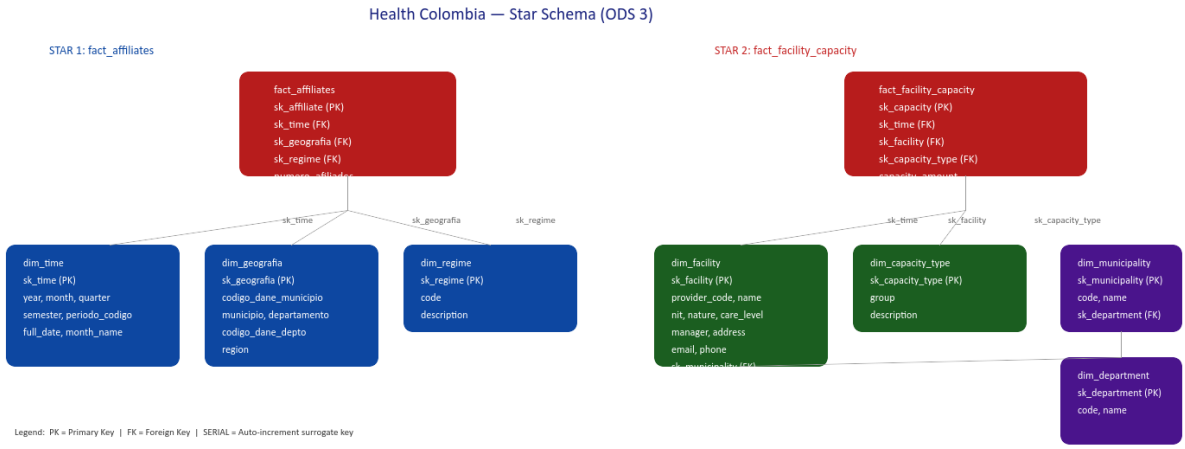


Figure 1: Star schema of the dimensional data warehouse.

5.2 Dimension Tables

Table	Justification	Records
`dim_time`	Temporal analysis required by R1, R2, R4, R5. Quarterly granularity.	2
`dim_geografia`	Geographic analysis required by R2, R4, R5. DANE codes + region.	2,240
`dim_department`	Department-level analysis required by R2, R4. Includes region.	34
`dim_municipality`	Municipality granularity for territorial equity analysis (R5).	1,168
`dim_regime`	Core dimension for R1, R5. Maps S/C/E/I to descriptive names.	4
`dim_facility`	Facility-level analysis for R3, R4. Provider attributes.	10,921
`dim_capacity_type`	Capacity types (CAMAS/SALAS × TPR/Adultos/-Pediátrica) for R3, R4.	63

Table 5: Dimension tables and their analytical justification.

5.3 Fact Tables

Fact Table	Grain	Measure	Records
`fact_affiliates`	Quarter + Municipality + Regime	`numero_afiliados`	3,369
`fact_facility_capacity`	Facility + Capacity Type + Time	`capacity_amount`	31,496

Table 6: Fact tables with grain and measures.

5.4 Why Two Fact Tables?

The two fact tables have **different granularities**:

- `fact_affiliates` is aggregated at (quarter, municipality, regime) — a many-to-many relationship between territory and regime.
- `fact_facility_capacity` is at (facility, capacity type, time) — individual facility-level infrastructure data.

These represent fundamentally different business processes (enrollment vs. infrastructure) and cannot be merged without losing analytical precision.

6 ETL Pipeline

6.1 Architecture

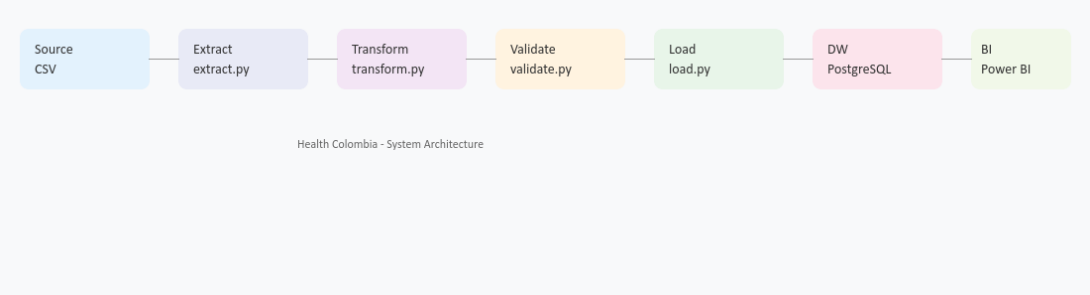


Figure 2: ETL pipeline architecture.

The pipeline follows a linear flow:

```
Source CSVs -> Extract -> Raw Validate -> Transform -> Validate -> Load ->
    PostgreSQL DW
                                                                        |
                                                                CSV Export ->
                                                                    data/
                                                                    processed/
```

6.2 Extract (`src/extract.py`)

- Reads raw CSV files with `pd.read_csv(dtype=str)` to preserve original formatting.
- Renames columns from Spanish to English for consistency.

6.3 Raw Validation (`src/validate.py`)

Validates the raw extracted data before any transformation:

- Verifies all required columns are present.
- Checks that `NumPersonas` contains no negative values.
- Validates regime codes are within {S, C, E, I}.

6.4 Transform (`src/transform.py`)

Data Preparation:

- Removes thousand-separator periods from `NumPersonas`, converts to integer.
- Normalizes department and municipality names (accent removal, uppercase, alias mapping).
- Maps departments to regions using `REGION_MAP` (after normalization, ensuring correct mapping).
- Maps "NO APLICA" to "SIN DEPARTAMENTO" with synthetic DANE code "00" (preserves 1,315,701 affiliates).
- Maps special health districts (Cali, Cartagena, etc.) to their real departments.
- Extracts numeric digits from phone numbers; removes commas from NIT values.

Dimensional Transformation:

- Builds 7 dimension tables with surrogate keys (`sk_*`).
- Generates deterministic DANE codes for facility-only municipalities using MD5 hashing.
- Aggregates `fact_facility_capacity` with `groupby().sum()` to eliminate duplicate capacity rows.

6.5 Validate (`src/validate.py`)

Rule	Description
Raw Validation	Validates raw data before transformation: required columns, no negative `NumPersonas`, valid regime codes (S/C/E/I)
FK Integrity	All foreign keys in fact tables reference existing dimension surrogate keys
Null Measures	No null values in `numero_afiliados` or `capacity_amount`
Negative Measures	No negative values in measure columns
Duplicate Keys	No duplicate composite keys in fact tables
Sum Consistency	Total `numero_afiliados` matches raw CSV total within 1% tolerance
Row Count	Unique facilities in fact table match source unique facilities

Table 7: ETL validation rules.

6.6 Load (`src/load.py`)

- Connects to PostgreSQL using `psycopg2`.
- Executes `sql/init.sql` to reset schema (idempotent DROP + CREATE).
- Loads dimensions first, then fact tables, using `ON CONFLICT DO NOTHING`.
- Resets serial sequences after load.
- Single transaction: commits on success, rolls back on failure.
- **CSV Export:** After PostgreSQL load, exports all tables as individual CSV files to `data/processed/` for Power BI ingestion.

7 Data Quality Issues and Fixes

During development, several critical data quality issues were identified and resolved:

Issue	Root Cause	Fix
5,072 facility rows silently dropped	Special health districts (Cali, Cartagena, Barranquilla, Santa Marta, Buenaventura) reported as "department" in REPS	`DISTRICT_TO_DEPT` mapping redirects to real departments
80 municipalities unmapped	CAUCA missing from `DEPT_DANE_CODES`; facility-only municipalities not in affiliates	Added CAUCA (code 19); `build_dim_municipality` merges both datasets
13,970 duplicate fact keys	Raw REPS has duplicate rows per facility + capacity type	`build_fact_facility_capacity` aggregates with `groupby().sum()`
1,315,701 affiliates lost ("NO APLICA")	`DEPT_NORMALIZE['NO APLICA'] = None` caused silent drop	Map to "SIN DEPARTAMENTO" with DANE code "00"
Cauca without region	CAUCA absent from `REGION_MAP`	Added `'CAUCA': 'Pacifico'`
Validation false positive	`len(df)` compared total rows vs unique facilities	Changed to `provider_code.nunique()`
Non-deterministic hash	Python `hash()` varies across runs	Replaced with `hashlib.md5()`
Raw total computed post-clean	Validation compared against already-cleaned total	`raw_total` computed before `clean_affiliates()`
29 facilities from AMAZONAS/GUAVIARE dropped	`DEPT_DANE_CODES` missing these departments	Added `'AMAZONAS': '91'`, `'GUAVIARE': '95'` + new `validate_department_consistency()`
Valle del Cauca (4.6M affiliates) mapped to "Sin Region"	`region` calculated before `DEPT_NORMALIZE` corrected "VALLE" → "VALLE DEL CAUCA"	Moved `region` calculation to `clean_affiliates()` after normalization; added `validate_department_region_consistency()`

Table 8: Data quality issues identified and resolved.

8 Analytical Queries and KPIs

All queries run against the PostgreSQL Data Warehouse. The complete SQL is in `sql/analytical_queries.sql`.

8.1 R1 — Total Affiliates by Regime Type

```
SELECT r.description AS regime,
      SUM(f.numero_afiliados) AS total_affiliates,
      ROUND(SUM(f.numero_afiliados) * 100.0
           / SUM(SUM(f.numero_afiliados)) OVER(), 2) AS pct_share
```

```
FROM fact_affiliates f
JOIN dim_regime r ON f.sk_regime = r.sk_regime
GROUP BY r.description
ORDER BY total_affiliates DESC;
```

Result: Subsidized and Contributory regimes are nearly balanced nationally (∼48% vs ∼47.5%). Special and Individual <5%. The real disparity emerges at the regional level (see R5).

8.2 R2 — Top and Bottom Departments

Uses UNION ALL to combine the top 10 and bottom 10 departments by total affiliate count.

Result: Bogotá D.C., Antioquia, and Valle del Cauca concentrate the largest populations. Vaupés, Guainía, and Vichada have the fewest.

8.3 R3 — Facilities by Nature and Care Level

```
SELECT fc.nature, fc.care_level,
      COUNT(DISTINCT fc.sk_facility) AS facility_count,
      SUM(c.capacity_amount) AS total_capacity
FROM fact_facility_capacity c
JOIN dim_facility fc ON c.sk_facility = fc.sk_facility
GROUP BY fc.nature, fc.care_level
ORDER BY fc.nature, fc.care_level;
```

Result: Public facilities dominate high-complexity care (levels 3–4); private facilities concentrate in lower-complexity (levels 1–2). A marginal "Mixta" category (15 facilities) also exists.

8.4 R4 — Installed Capacity per Affiliate by Department

Uses CTEs with FULL OUTER JOIN to combine capacity and affiliation data by department.

Result: Smaller departments show highly variable ratios; some have excess capacity while others face critical shortages. SIN DEPARTAMENTO (non-geolocalized records) is excluded from this analysis.

8.5 R5 — Regime Distribution by Geographic Region

```
SELECT g.region, r.description AS regime,
      SUM(f.numero_afiliados) AS total_affiliates,
      ROUND(SUM(f.numero_afiliados) * 100.0
            / SUM(SUM(f.numero_afiliados))
              OVER(PARTITION BY g.region), 2) AS pct_within_region
FROM fact_affiliates f
JOIN dim_geografia g ON f.sk_geografia = g.sk_geografia
JOIN dim_regime r ON f.sk_regime = r.sk_regime
GROUP BY g.region, r.description
ORDER BY g.region, total_affiliates DESC;
```

Result: Andina region concentrates the majority. Pacifico shows a strong Subsidized majority (driven by Valle del Cauca, Chocó, and Nariño). Amazonia has the highest proportional Subsidized concentration.

9 Validation Results

After the full ETL pipeline execution, the following results confirm data integrity:

9.1 Load Summary

Table	Records
dim_time	2
dim_geografia	2,240
dim_department	34
dim_municipality	1,168
dim_regime	4
dim_facility	10,921
dim_capacity_type	63
fact_affiliates	3,369
fact_facility_capacity	31,496

Table 9: Final load summary.

9.2 Data Quality Checks

Check	Result
FK nulls (fact_affiliates)	0
FK nulls (fact_facility_capacity)	0
Negative measures	0
Duplicate fact keys	0
Sum consistency (affiliates)	51,182,238 = 51,182,238 (100% match)
Unmapped municipalities	0
Total validation errors	**0**

Table 10: Data quality verification results.

10 Business Intelligence

A Power BI dashboard connects to the PostgreSQL Data Warehouse and provides:

- **Geographic Analysis:** Department-level map visualization showing affiliate density and capacity distribution.
- **Comparative Analysis:** Public vs. private facility distribution across care levels.
- **KPIs:** Total affiliates nationally, total installed capacity, capacity-to-affiliate ratio.
- **Filters:** By department, region, regime type, facility nature, and time period.

Data Loading Options for Power BI:

1. **PostgreSQL connection** (recommended): Direct connection using the PostgreSQL connector.
2. **CSV import:** Load individual CSV files from `data/processed/` (each includes surrogate keys for joining).



Figure 3: Power BI dashboard overview.

11 Technologies

Component	Technology
Programming	Python 3.12, Pandas, NumPy
Profiling	Jupyter Notebook
Data Warehouse	PostgreSQL 16
ETL Pipeline	Python (extract → transform → validate → load)
BI Dashboard	Power BI
Containerization	Docker / Podman
Version Control	Git, GitHub
Development Env	Nix Flake

Table 11: Technology stack.

12 Limitations and Assumptions

1. **Temporal snapshot:** Both datasets are cross-sectional (Q2 2022 affiliates, Q4 2022 facilities). True trend analysis requires multiple periods.
2. **Care level gaps:** 61% of facility records lack a care level. Missing values stored as NULL in the data warehouse.
3. **Affiliation ≠ Access:** High affiliation rates do not guarantee effective healthcare access. The analysis measures enrollment, not utilization.

4. **Capacity definition:** `capacity_amount` represents installed capacity, not operational or effective capacity.
5. **Cross-dataset caveat:** R4 uses Q4 2022 capacity with Q2 2022 affiliation. The ratio is indicative, not temporally precise.
6. **Facility municipalities:** Facilities in municipalities absent from the affiliates dataset are recovered via synthetic DANE codes (MD5 hashing).

13 Conclusions

This project demonstrates the complete lifecycle of an ETL process applied to Colombian healthcare data:

1. **Data integration:** Two complementary datasets (SISPRO affiliates + REPS facilities) were integrated into a unified dimensional model with 7 dimension tables and 2 fact tables.
2. **Data quality:** Eleven critical data quality issues were identified and resolved, including silent data loss from special health districts, missing department codes, duplicate capacity rows, the "NO APLICA" gap, and the Valle del Cauca region mapping error.
3. **Analytical coverage:** All five analytical requirements (R1–R5) are fully supported by the dimensional model and verified through SQL queries.
4. **Zero data loss:** Final validation confirms 100% sum consistency (51,182,238 affiliates), 0 FK violations, 0 duplicate keys, and 0 negative measures.
5. **BI readiness:** The data warehouse is accessible via PostgreSQL connection or CSV export for Power BI dashboard creation.

The dimensional data warehouse provides a solid foundation for healthcare analytics aligned with SDG Target 3.8, enabling stakeholders to identify geographic disparities in affiliation coverage and infrastructure capacity across Colombia's territories.

Team Members

Member	Role	Responsibilities
Deyton Riascos Ortiz	Project Manager	Coordinate the project, organize tasks, consolidate the final report
Samuel Izquierdo Bonilla	Development Team	Lead requirements definition, user stories, KPIs, data mapping
Daniel David Garcia Restrepo	Product Owner	Prioritize requirements, validate client needs, review business objectives
Mauricio Taborda Gongora	Tester	Validate data quality, verify ETL results, perform testing

Table 12: Team members and their roles.